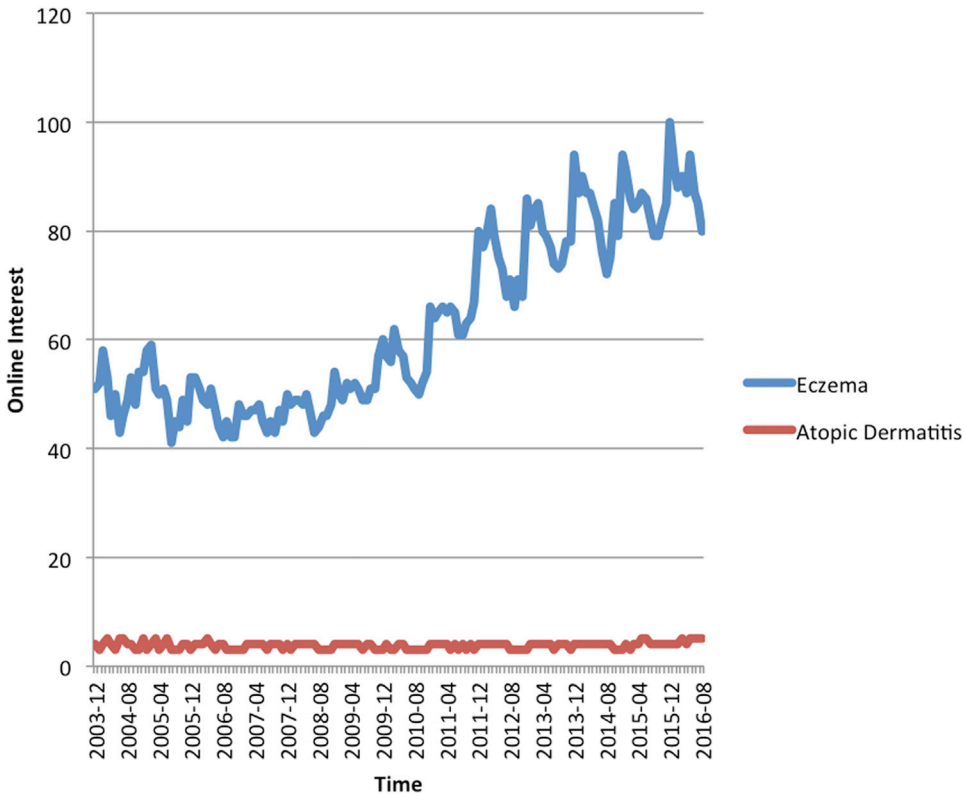

Fig 1. Relative Google search interest for "eczema" and "atopic dermatitis" from December 2003 through August 2016. The y-axis depicts search interest relative to the highest point on the chart (a value of 100 represents peak popularity for the search term, while a value of 50 means that the term is half as popular). Data source: Google Trends (www.google.com/trends).[4]

Table I. Number of websites containing information consistent with the American Academy of Dermatology guidelines, by search term used

Search term	Consistent, n (%)	Inconsistent, n (%)	Not relevant, n (%)
Eczema treatment	13 (32.5)	18 (45.0)	9 (22.5)
Atopic dermatitis treatment	12 (30.0)	17 (42.5)	11 (27.5)
Overall	25 (31.3)	35 (43.8)	20 (25.0)

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http://dx.doi.org/10.1016/j.jaad.2017.01.054

Itch intensity in moderate-to-severe plaque psoriasis versus atopic dermatitis: A meta-analysis

To the Editor: Atopic dermatitis is commonly referred to as the "itch that rashes,"[1] whereas plaque psoriasis has not traditionally been viewed as a pruritic dermatosis. More recently, the role of pruritus in psoriasis has emerged as an important symptom affecting quality of life, and clinical trials for psoriasis have begun using itch scores more consistently as an outcome measure.[2,3] To date, there have been no studies directly comparing itch intensity in atopic dermatitis and plaque psoriasis. This meta-analysis, performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement from the Cochrane Collaboration guidelines, used pooled data from randomized, clinical trials to compare mean baseline pruritus scores in these 2 diseases.

An electronic search was conducted in PubMed, using the terms "atopic dermatitis or psoriasis," a list of systemic therapies used to treat these conditions, and "itch or pruritus." In addition, filters limiting the results to "clinical trials" and "humans" were placed. Relevant publications were also found by searching reference lists. Articles were selected using the following inclusion criteria: randomized trials using systemic therapies to treat adults with moderate to severe plaque psoriasis or atopic dermatitis, and available baseline pruritus data on a 10- or 100-point numerical rating scale or visual analogue scale.

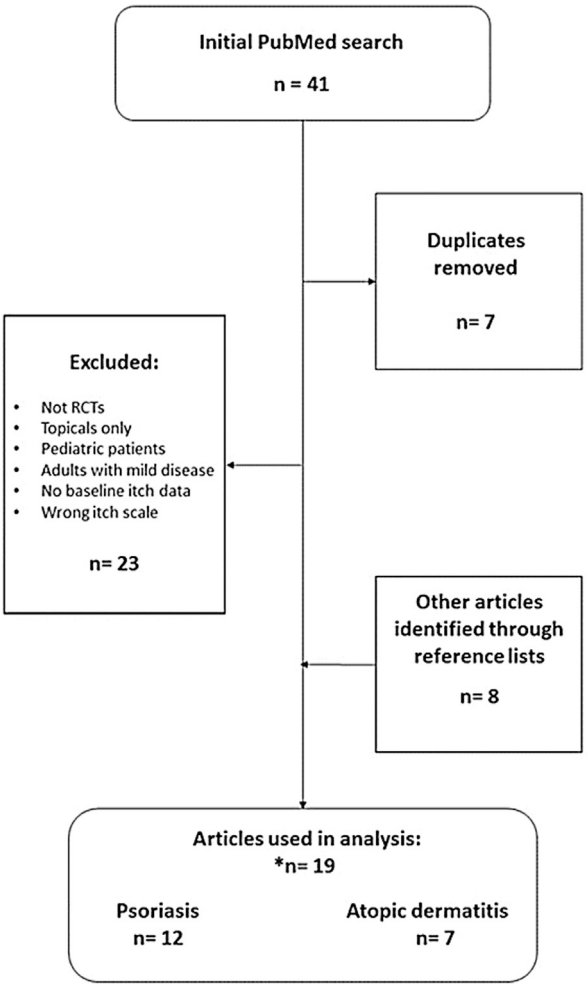

Fig 1. Study selection flow diagram. *RCTs*, Randomized, controlled trials; *n*, number of articles. *A total of 22 RCTs (13 psoriasis, 9 atopic dermatitis) were presented in 19 articles (2 of the articles included data from multiple trials).

Baseline pruritus data were abstracted from each article. If a 10-point scale was used, all values were multiplied by 10 to allow for synthesis of data on a 100-point scale. The data were weighted on the basis of the number of subjects in each study arm, and overall mean itch scores for atopic dermatitis and psoriasis were calculated. The 95% confidence intervals were constructed around the weighted means, using an α of 0.05 and n values equal to the number of trials included in each arm, and statistical differences were considered to be significant when there was no overlap between the 2 intervals.

Forty-one articles were identified from the initial search. Seven articles with duplicate data were removed, 23 articles were removed according to the exclusion criteria, and 8 additional articles were identified through reference lists. This resulted in a total of 19 included articles, which presented data from 22 trials (Fig 1 and Supplemental Table I, available at http://www.jaad.org). The weighted mean baseline pruritus scores were 63.03 for psoriasis and 62.87 for atopic dermatitis. There was no statistically significant difference between the 2 groups (Fig 2).

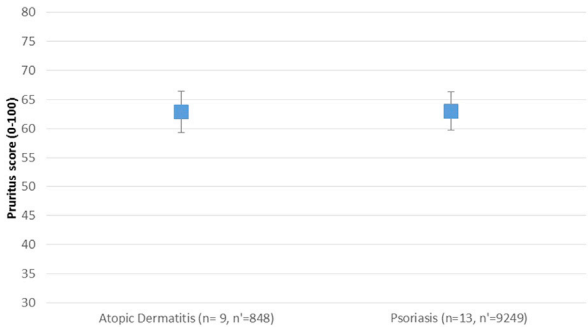

Fig 2. Weighted mean baseline pruritus scores in patients with psoriasis and atopic dermatitis with 95% confidence intervals. *n*, Number of trials; *n'*, number of subjects.

In conclusion, although psoriasis is traditionally viewed as less pruritic than atopic dermatitis, there is no significant difference in mean baseline itch scores in patients warranting systemic treatment. Pruritus levels probably are affected by a variety of confounding factors; however, this does suggest that overall, itch is a more significant component of psoriasis than previously recognized. Pruritus in psoriasis has been associated with occupational impairment,[4] anxiety, depression, and a negative impact on overall quality of life, mood, concentration, sleep, sexual desire, and appetite.[5] Pruritus may also worsen psoriatic disease by increasing scratching and subsequent koebnerization.

Fortunately, an improvement in Psoriasis Area and Severity Index score with treatment has been shown to correlate with reduced itch severity and improved quality of life.[2] Further research is needed to elucidate the relationship between psoriasis and pruritus. In addition, pruritus should continue to be included as an outcome measure in clinical trials, as it appears to have a substantial impact on treatment satisfaction and quality of life.

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Analysis of US dermatology physician assistant density

To the Editor: The density of dermatologists in the United States has been slowly increasing over the past decade, but a material geographic variance exists, leaving many regions underserved.[1] Patients with skin disorders may often experience long wait times because the average dermatologist density remains below the recommended 4 per 100,000 population.[1,2] Physician assistants have become a fixture of contemporary medicine and their role within dermatology has expanded dramatically in the past decade.[3] The purpose of this study was to examine the density of dermatology physician assistants (DPAs) throughout the United States to determine how their distribution may extend patient options for dermatologic care.

A cross-sectional analysis was performed using membership data obtained from the Society of Dermatology Physician Assistants (SDPA) on the number of practicing DPAs in the United States in 2016 by ZIP code. SDPA membership represents >75% of US DPAs.[4] DPA to population ratios were calculated using data obtained from the US Census Bureau for 3-digit ZIP code locations (section codes). Adjacent section codes were combined where appropriate.

Table I. Most dermatology physician assistant–dense areas in the United States

Rank	3-digit postal code	Location	No. of DPAs	DPAs per 100,000 population
1	830	Rock Springs, WY	3	14.5
2	022	Boston, MA (downtown)	2	7.1
3	185	Scranton, PA	7	5.9
4	214	Annapolis, MD	5	5.4
5	507	Waterloo, IA	4	5.1
6	803	Boulder, CO	6	5.0
7	101	Manhattan, NY (upper east side)	3	4.8
8	502	Des Moines, IA	10	4.7
9	253	Charleston, WV	5	4.3
10	998	Juneau, AK	2	4.0
11	915	Burbank, CA	4	3.7
12	571	Sioux Falls, SD	6	3.7
13	271	Winston Salem, NC	9	3.6
14	685	Lincoln, NE	9	3.4
15	288	Asheville, NC	4	3.3
16	235	Norfolk, VA	8	3.3
17	734	Ardmore, OK	3	3.2
18	844	Ogden, UT	6	3.2
19	534	Racine, WI	4	3.1
20	837	Boise, ID	8	3.1
21	591	Billings, MT	4	3.1
22	897	Carson City, NV	2	3.0
23	260	Wheeling, WV	4	3.0
24	181	Allentown, PA	5	3.0
25	268	Petersburg, WV	1	3.0

DPA, Dermatology physician assistant.

In 2016, there were 2520 US SDPA DPAs. The US density of DPAs was 0.78 per 100,000 persons. The 25 densest DPA locations in 2016 are summarized in Table I.

Dermatologists are unevenly geographically distributed, with most typically practicing in major urban areas or large academic centers.[1] The supply of new dermatologists is limited by the number of residency training slots, so provider growth has been primarily from the addition of DPAs. Combining the 0.78 DPAs per 100,000 in the United States with the 3.4 dermatologists per 100,000 increases the dermatology provider density to >4 per 100,000, the number needed to adequately care for a population (if the geographic distribution was uniform).[1]

The combination of US dermatologists with the substantial influx of DPAs is correcting the undersupply and maldistribution to some extent. Although the density of DPAs is also not uniform, most

Supplemental Table I. Reference list of articles included in the meta-analysis

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